

Sandeep Birudukota

Data Engineer

sandeepbirudukota@jobemails.net | (669)-204-4373 | San Jose, CA | LinkedIn | GitHub

Summary

Data Engineer with over 4+ years of experience in building scalable data pipelines and architectures. Proficient in Python and SQL, with expertise in ETL processes using Apache Airflow and Talend. Experienced in data warehousing with Amazon Redshift and Google BigQuery, as well as big data frameworks like Spark and Hadoop. Skilled in cloud platforms such as AWS and Azure, and strong in automation using Terraform. Known for effective communication and problem-solving abilities to ensure high-quality project delivery.

Technical Skills

- **Programming Languages:** Python, SQL, NoSQL, Shell, R, HTML, CSS
- **Frameworks & Tools:** NumPy, Pandas, Matplotlib, Flask, Django, Git, Docker, CI/CD, Statistics, Agile, SAS, Hadoop, Hive, PySpark, Snowflake, Apache (Airflow, Kafka), Zookeeper, Linear Regression, Decision Trees, Random Forest, Terraform, Databricks
- **Databases:** MS-SQL, MySQL, Redshift, Snowflake, Big Query, PostgreSQL, MongoDB, SparkSQL, Oracle, Azure, AWS
- **Data Analysis / ETL:** Power BI, Tableau, Sterling Integrator, Databricks, MS Excel, Google Analytics, Machine Learning, Data Modeling, Data Mapping, Data Mining, Data Extraction, Transformation, Informatica

Professional Experience

Data Engineer, Deloitte

03/2024 – Present | Remote, USA

- Developed a scalable data lake on GCP using Cloud Storage to store over 5 TB of diverse datasets, for efficient retrieval and analysis of structured and unstructured data across multiple projects within the organization.
- Implemented data ingestion pipelines utilizing Apache Kafka with Zookeeper for real-time data streams, processing over 10,000 events per second, alongside Hadoop for batch processing.
- Leveraged PySpark within Databricks to run distributed data processing jobs, using Pandas for data manipulation and optimization, for high-quality datasets were available for downstream analytics and machine learning applications, processing 2 million records daily.
- Employed Hive for SQL-based querying of large datasets, utilizing advanced SQL techniques like JOINS and window functions to derive insights, improving reporting capabilities and reducing query time by 40%.
- Established a data governance framework using HBase for efficient metadata management and data discovery, improving compliance and facilitating easier access to data for over 50 analysts and data scientists.
- United with teams to provide clean and processed data for analysts, enabling them to create impactful dashboards in Power BI, leading to a 30% gain in data-driven decision-making across the organization.

Data Engineer, USAA

05/2023 – 02/2024 | USA

- Enhanced AWS infrastructure by automating resource provisioning using Infrastructure as Code (IaC), resulting in a 50% drop-in deployment time and significantly improving overall operational efficiency.
- Designed an end-to-end streaming pipeline utilizing Apache Spark and Kafka, which enabled the processing of 10,000 events per minute, with PySpark facilitating streamlined data transformations for real-time applications.
- Integrated various AWS services, including S3 for data storage, AWS Glue for ETL processes, and Amazon Redshift for data warehousing, using CTEs in SQL to enhance the readability and maintainability of complex queries.
- Developed PySpark scripts in AWS Glue to reshape schemas and enrich data. Additionally, using Pandas to efficiently manipulate and clean datasets before loading into databases.
- Leveraged AWS Glue and AWS Athena for data transformations and ad-hoc querying, providing stakeholders with real-time insights. SQL techniques were employed to effectively query data stored in S3.
- Streamlined the deployment process via Docker integration and empowered data-driven decisions by providing clean, transformed data to analysts for use in Power BI, enabling the creation of dynamic visualizations.

Data Engineer, TATA Consultancy Services	09/2019 – 12/2021 India
- Designed an ETL pipeline to migrate data from SQL Server to Azure Data Lake, enhancing data integration and bolstering security measures for sensitive business data while ensuring compliance with regulatory needs. - Used Azure Data Factory for data ingestion, employing Data Flows and Triggers for automation, achieving a 60% boost in processing speed and ensured optimal storage in Parquet format for improved data accessibility. - Leveraged Azure Databricks, PySpark, and Pandas for complex data transformations, resulting in a 50% drop-in processing time enabling faster insights, improving decision-making abilities for various stakeholders. - Improved query performance by strategically loading data into Azure Synapse and optimizing SQL queries for large datasets exceeding 500 GB, easing response times and operational efficiency for analytics. - Executed partitioning strategies in Azure Synapse using SQL indexing techniques, achieving a 50% gain in data retrieval and analytical speed. - Architected a robust data warehouse framework focused on denormalization for performance optimization, facilitating efficient business intelligence activities. - Implemented rigorous validation and cleansing techniques within Azure Data Factory to enhance data quality, reducing discrepancies and increased the reliability of the data used for analysis.	

Data Engineer, NR IT Software Services	04/2019 – 09/2019 India
- Developed real-time ETL pipelines for retail client data using Hadoop and Hive for batch processing, improving data reliability by 40%, for timely insights that enabled informed decision-making for over 100 stakeholders. - Enforced AWS EventBridge to trigger AWS Lambda functions hourly, automating data retrieval from retail APIs, and use Kafka with Zookeeper for real-time data streaming, enhancing data consistency for 50+ data sources. - Formed Snowpipe with SQS for automatic ingestion into Snowflake, and use PySpark for efficient data transformations, provide quicker access to critical data, easing analysis of over 1 TB of data for reporting. - Adopted Databricks to streamline collaborative data processing workflows, managing complex datasets to improved coordination among 10+ team members and accelerated overall data analysis processes. - Provided clean, structured data to analysts for developing over 10 Power BI dashboards. - Observed infrastructure with AWS CloudWatch and implemented HBase for real-time access to 500,000 records.	

Education	
Master of Science, San Jose State University Software Engineering	01/2022 – 12/2023 San Jose, USA
Bachelor of Commerce, Andhra University Computer Science Engineering	08/2015 – 04/2019 AP, India